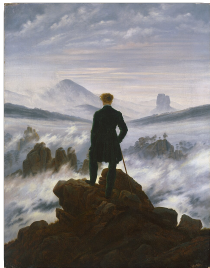


The Island of Truth

Immanuel Kant, *Critique of Pure Reason* (1781)

A land of truth in a sea of illusion



C. D. Friedrich,

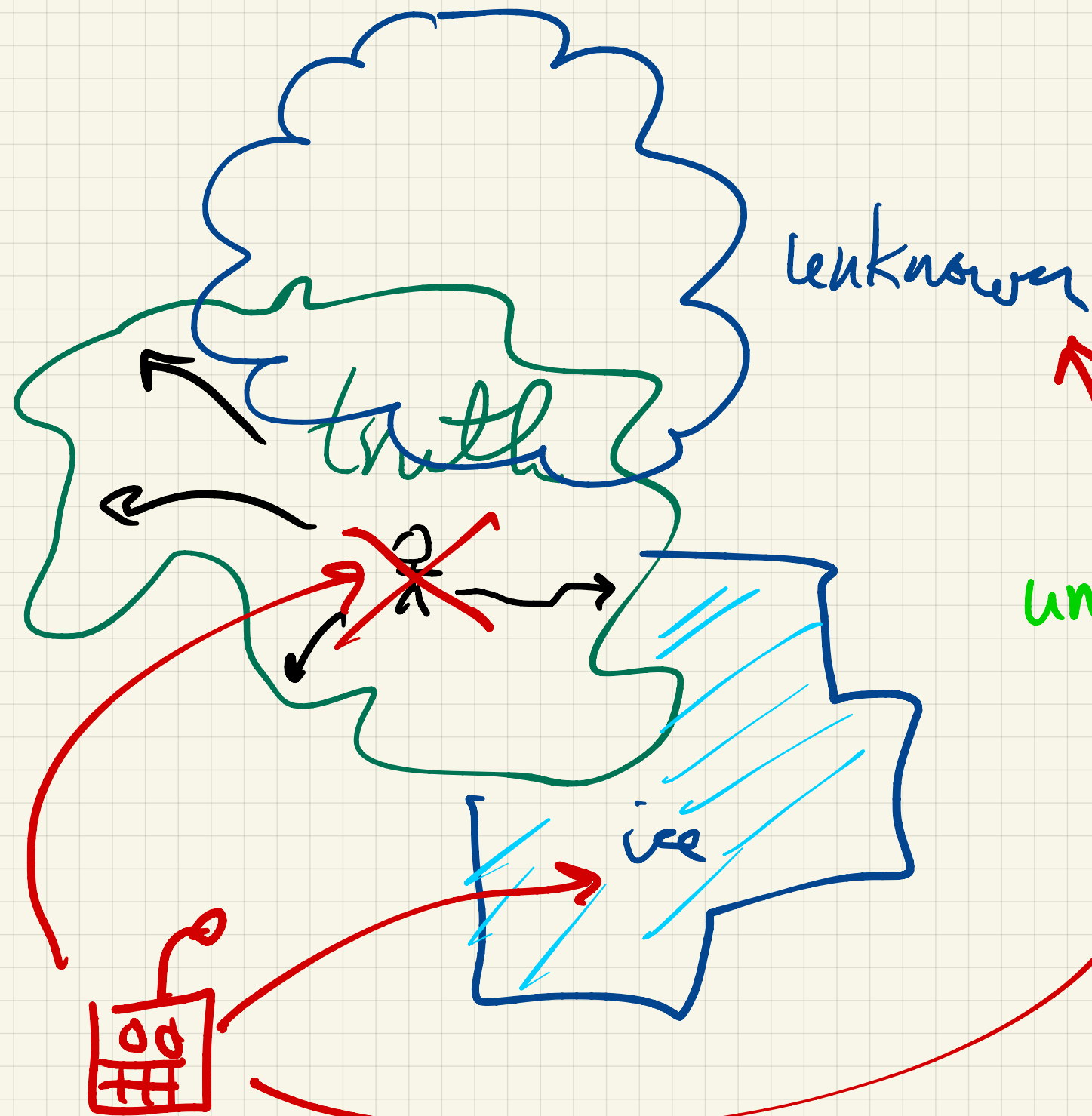
Wanderer above the Sea of Fog (c. 1817)

Es ist das **Land der Wahrheit** (ein reizender Name), umgeben von einem weiten und stürmischen Oceane, dem **eigentlichen Sitze des Scheins**, wo manche **Nebelbank** und manches bald wegschmelzende Eis **neue Länder lügt** ...

It is the *land of truth* (an enchanting name), surrounded by a wide and stormy ocean, the very *seat of illusion*, where many a *fog bank* and many a swiftly melting iceberg *feign new lands* ...

Kritik der reinen Vernunft, A235 / B294 (1781 / 1787)

To seek the truth is to know the fog for fog, and never chart land that is not there.



Weekend 1

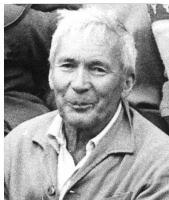
1) Recognize & quantify the

"fog"

2) Recognize & "avoid" "ice"

Weekend 2

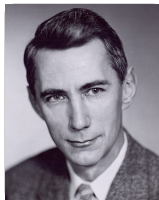
Two ways to measure the fog



Andrey Kolmogorov (1903 to 1987)

Probability theory

variance: how far outcomes spread around the average



Claude Shannon (1916 to 2001)

Information theory

entropy: the average surprise, in yes/no questions

Entropy and variance: two closely linked measures of the same fog.

The formulas behind the weekend

Information theory (Shannon)

$$H(X) = - \sum_x p(x) \log_2 p(x)$$

entropy: the average surprise, in bits

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

mutual information: what X tells you about Y

$$D_{\text{KL}}(p \parallel q) = \sum_x p(x) \log_2 \frac{p(x)}{q(x)}$$

relative entropy: the cost of believing q , not p

Probability theory (Kolmogorov)

$$\text{Var}\left(\sum_i X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

spread of a total: the spreads, plus how the parts move together

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} + \left(1 - \frac{1}{n}\right) \bar{\rho} \sigma^2$$

averaging shrinks the spread, unless the parts agree

Roadmap of the week

